Workshop Pole Barn: Sections and Construction Details

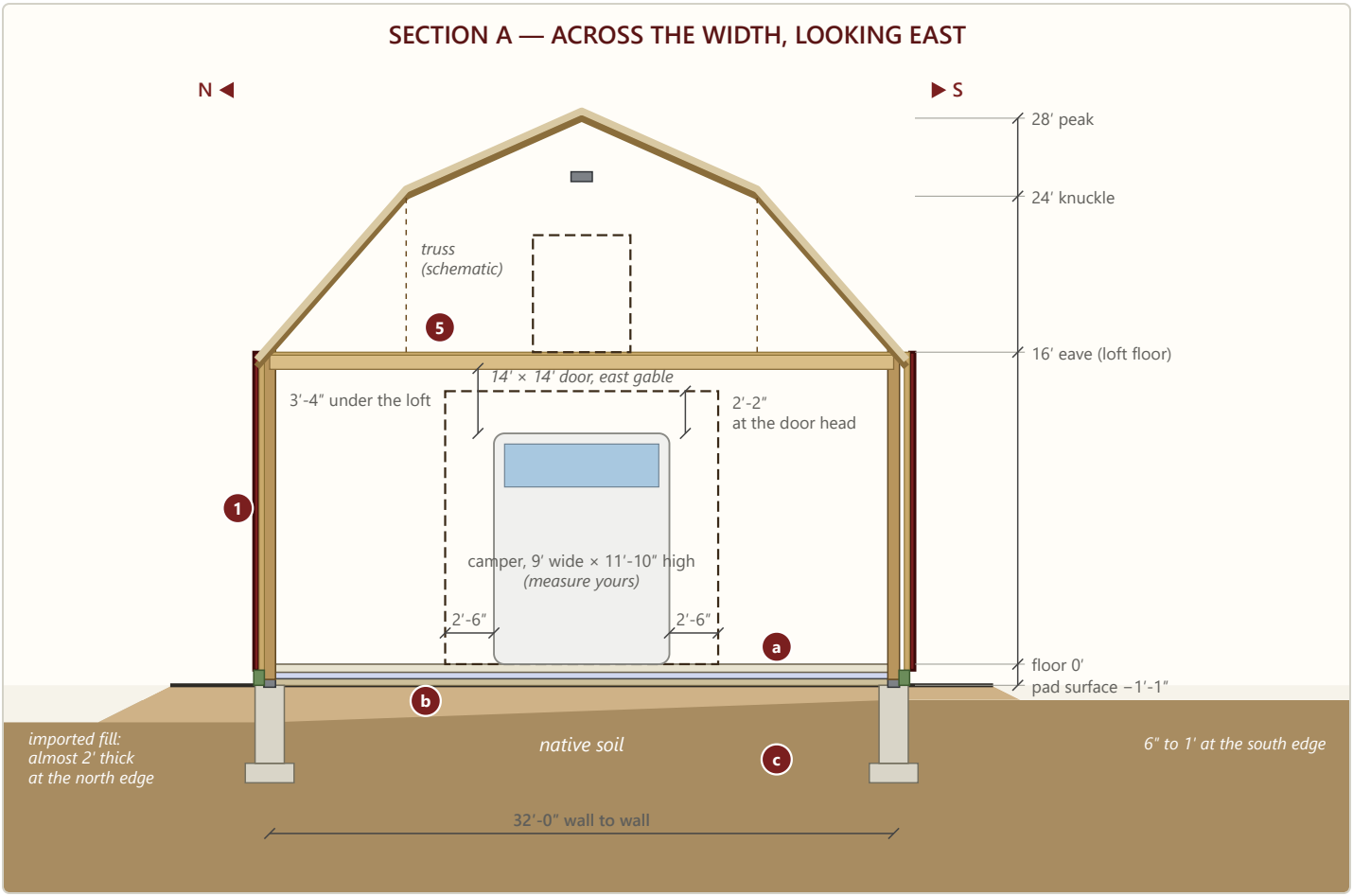
Cross-sections through the building, the two framing plans, and seven layer-by-layer details · 32 × 48 ft · 16 ft eave · gambrel

These are design-intent drawings for the engineer, the truss plant, and the builder. Truss members and connections are placeholders until the truss designer and the engineer size them. Layers in the details are exaggerated where they say so. The markers match the tables under each drawing.

What Is in This Section

1. **Section A, across the width**, looking east: the gambrel truss, the loft floor, the camper in the door, and the pad's fill under the slab.
2. **Section B, along the length**, looking north: the loft floor, the stairs, the main door, the hoist beam, and the deck.
3. **Plan F1, foundation and base framing**: piers, posts, the splash board, and the slab joints, on the pad.
4. **Plan F2, loft floor framing**: the 25 truss bottom chords at 2′ OC, the stair opening, the loft door header, the hoist beam, and the deck.
5. **Detail 1** wall assembly · **2** roof assembly · **3** pier and post base · **4** rainscreen base · **5** eave vent-out · **6** loft door, hoist beam, and deck · **7** truss bearing.

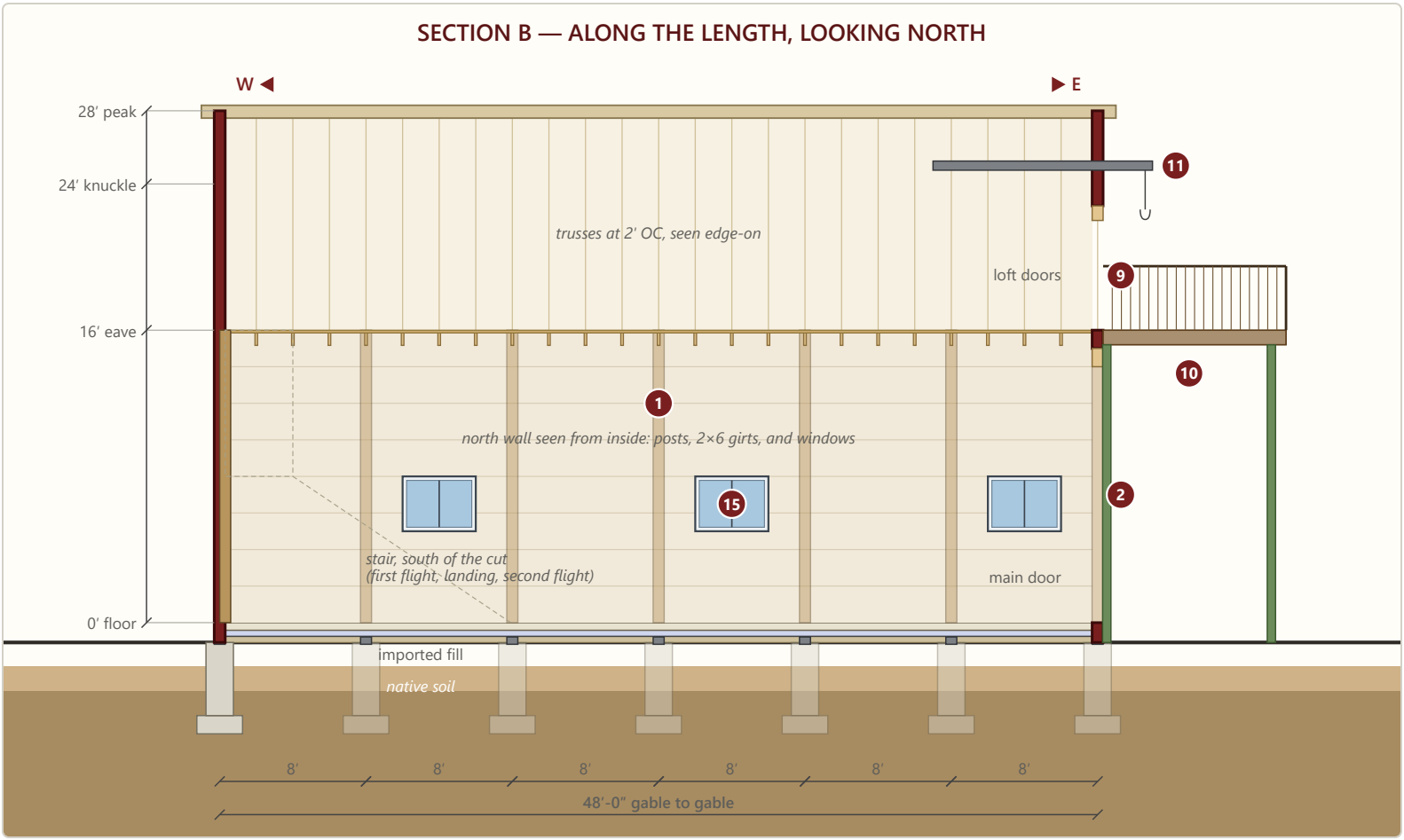
Section A: Across the Width (looking east)



Section A cuts across the 32 ft width and looks east at the gable wall. The 14′ × 14′ door and the loft doors on the east gable are drawn dashed behind the cut. The camper has 2′-2″ under the door head and 3′-4″ under the loft with an 11′-10″ roof; measure yours. The truss is a schematic envelope.

Marker	What it shows
1	3-ply 2×8 laminated post on its pier, with the wall layers outside it (steel, rainscreen, sheathing) and the splash board at the base
5	¾″ T&G plywood loft floor on the truss bottom chords, 16′ above the floor
a	The floor stack: 5″ slab on 4″ rigid foam and a 10 mil vapor barrier on 4″ compacted gravel, about 1′-1″ above the pad surface
b	Imported fill, the level pad: almost 2′ thick at the north edge and 6″ to 1′ at the south edge. The shoulders are drawn at 2:1 and are not surveyed
c	Native soil. The pier pads bear here, about 3′ down into it at the north row and 4′ to 4½′ at the south row

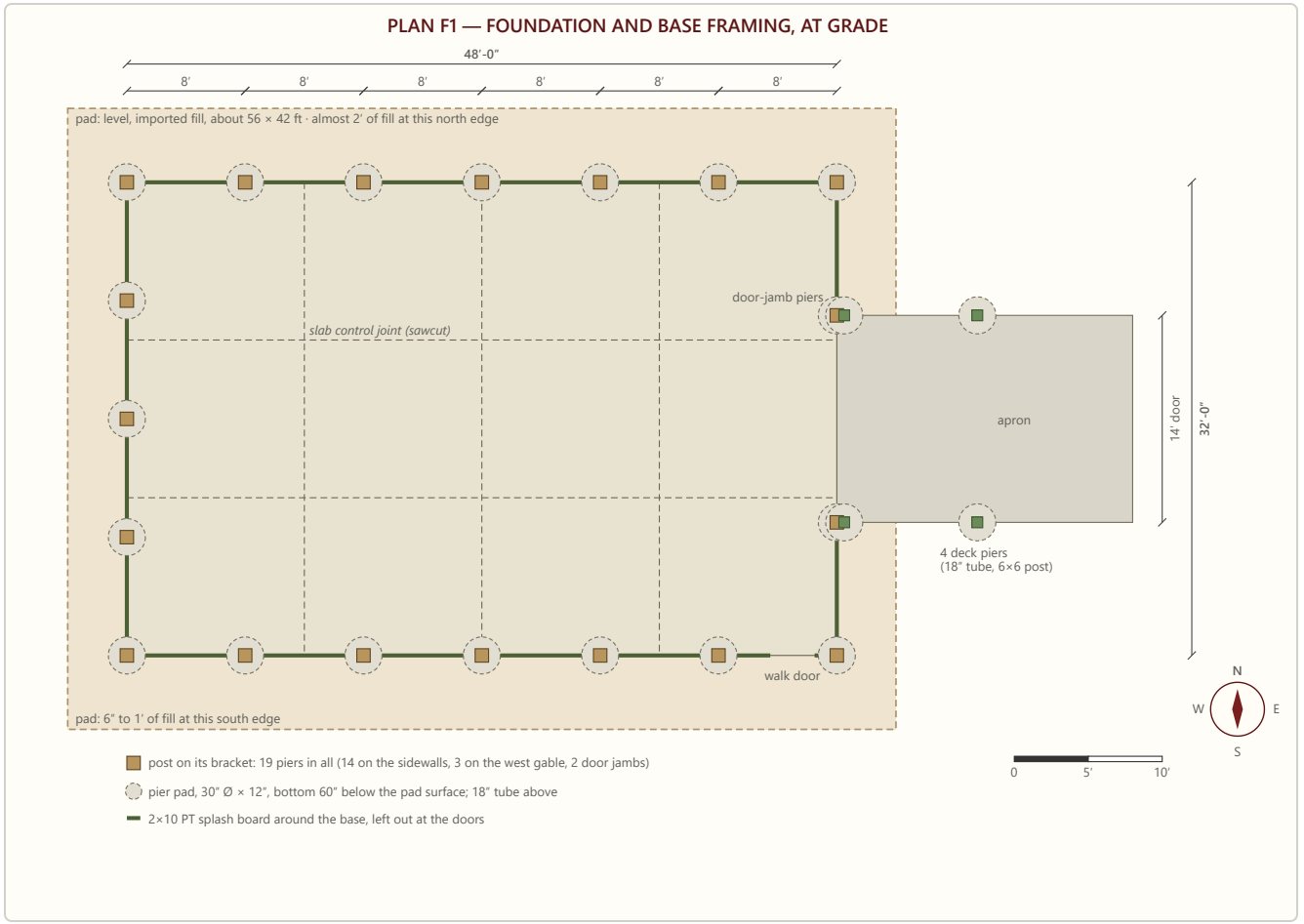
Section B: Along the Length (looking north)



Section B cuts along the 48 ft length at the center line and looks north. Sidewall posts are at 8-8-8-8-8-8 ft; the trusses (2′ OC) are seen edge-on; the loft floor and the ridge are cut. The three windows in the north wall show between the posts. The east gable shows the main door, the loft doors, the hoist beam, and the deck. The stair is south of the cut, so it is drawn dashed.

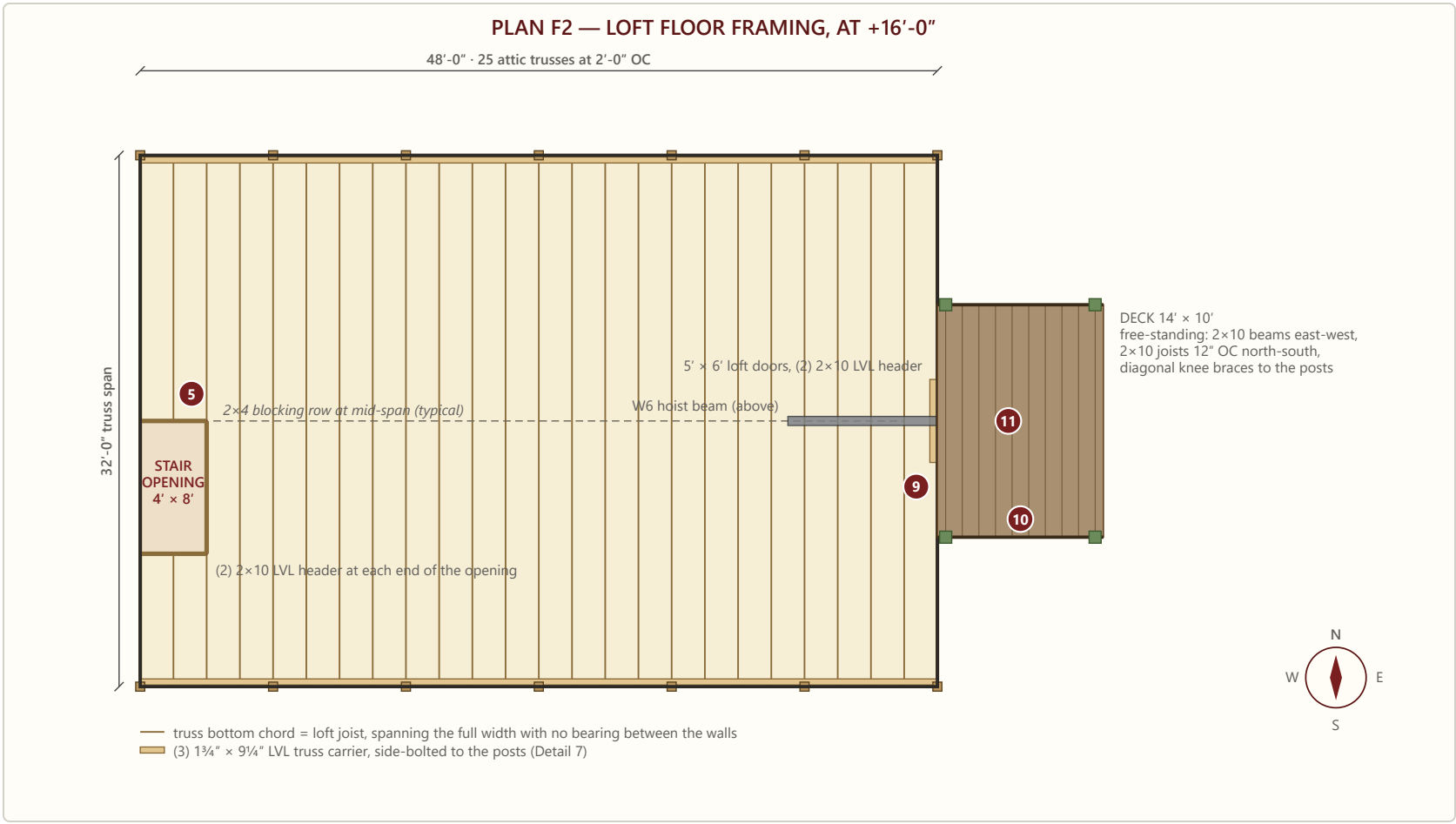
Marker	What it shows
1	Sidewall post (the north wall seen from inside, with its 2×6 girts)
2	The 14′ × 14′ main door opening in the east gable, with its LVL header above
9	The 5′ × 6′ loft doors at the loft floor, with their header
10	The 14′ × 10′ landing deck, 10′ deep, on posts to grade
11	The W6 hoist beam through the east wall, about 3 ft out and 9 ft in, with the chain fall at its tip
15	The north wall's windows, 4′ × 3′, sill 5′ above the floor, centered between the posts in bays 2, 4, and 6

Plan F1: Foundation and Base Framing



Plan F1, at grade. **19 building piers** (14 on the sidewalls at 8-8-8-8-8-8, 3 on the west gable, 2 door-jamb piers) plus 4 deck piers. All are 30″ Ø × 12″ pads, bottom 60″ below the pad surface.

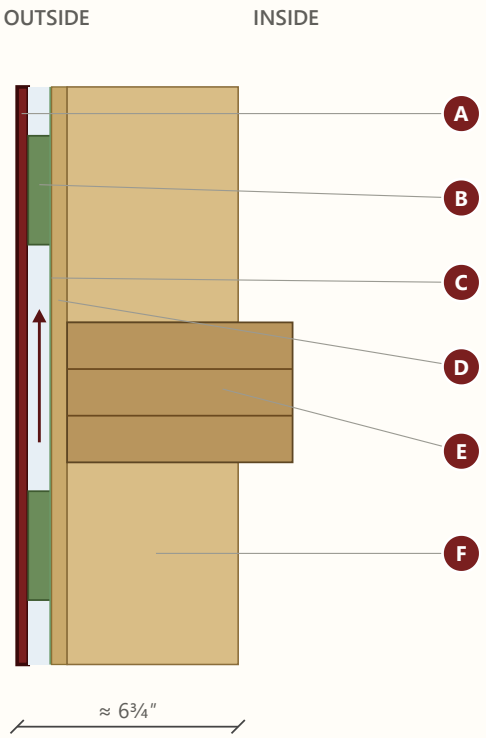
Plan F2: Loft Floor Framing



Plan F2, at +16′-0″. **25 attic-truss bottom chords at 2′ OC act as the loft joists**, spanning the full 32 ft width with no bearing between the walls. Floor build-up: ¾″ T&G plywood, glued and screwed, its long edge across the chords; foam or insulation between the chords is optional because the loft is conditioned on both sides.

Detail 1: Wall Assembly

DETAIL 1 — WALL ASSEMBLY (horizontal section, layers exaggerated)



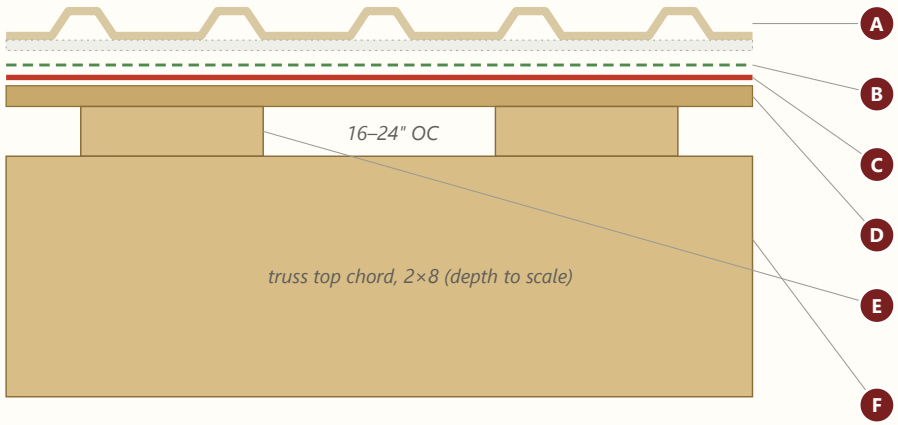
The furring stands off the sheathing to make a ¾″ air space that drains and dries (arrow: air moving up).

Detail 1, a horizontal slice through the wall at a girt, layers exaggerated. The ¾″ furring cavity is the most important detail in the wall: it lets the wall dry behind the steel and keeps moisture from reaching the sheathing.

Marker	Layer
A	26 ga steel siding, vertical panels, barn red, fastened to the furring with #10 × 1½″ hex screws with EPDM washers
B	1×4 PT furring at 24″ OC, vertical: makes a ¾″ ventilated rainscreen cavity
C	Tyvek CommercialWrap, taped seams, shingle-lapped, fastened with cap nails
D	½″ OSB sheathing, APA Exposure 1, 8d ring-shank nails at 6″ on the edges and 12″ in the field (4″ at corners and openings)
E	3-ply 2×8 laminated PT post at 8′ OC, on a wet-set bracket (Detail 3)
F	2×6 SPF #2 girt, bookshelf orientation (5½″ deep), 24″ OC, hung or notched to the posts

Detail 2: Roof Assembly

DETAIL 2 — ROOF ASSEMBLY (section at a purlin, lower slope, shown flat)

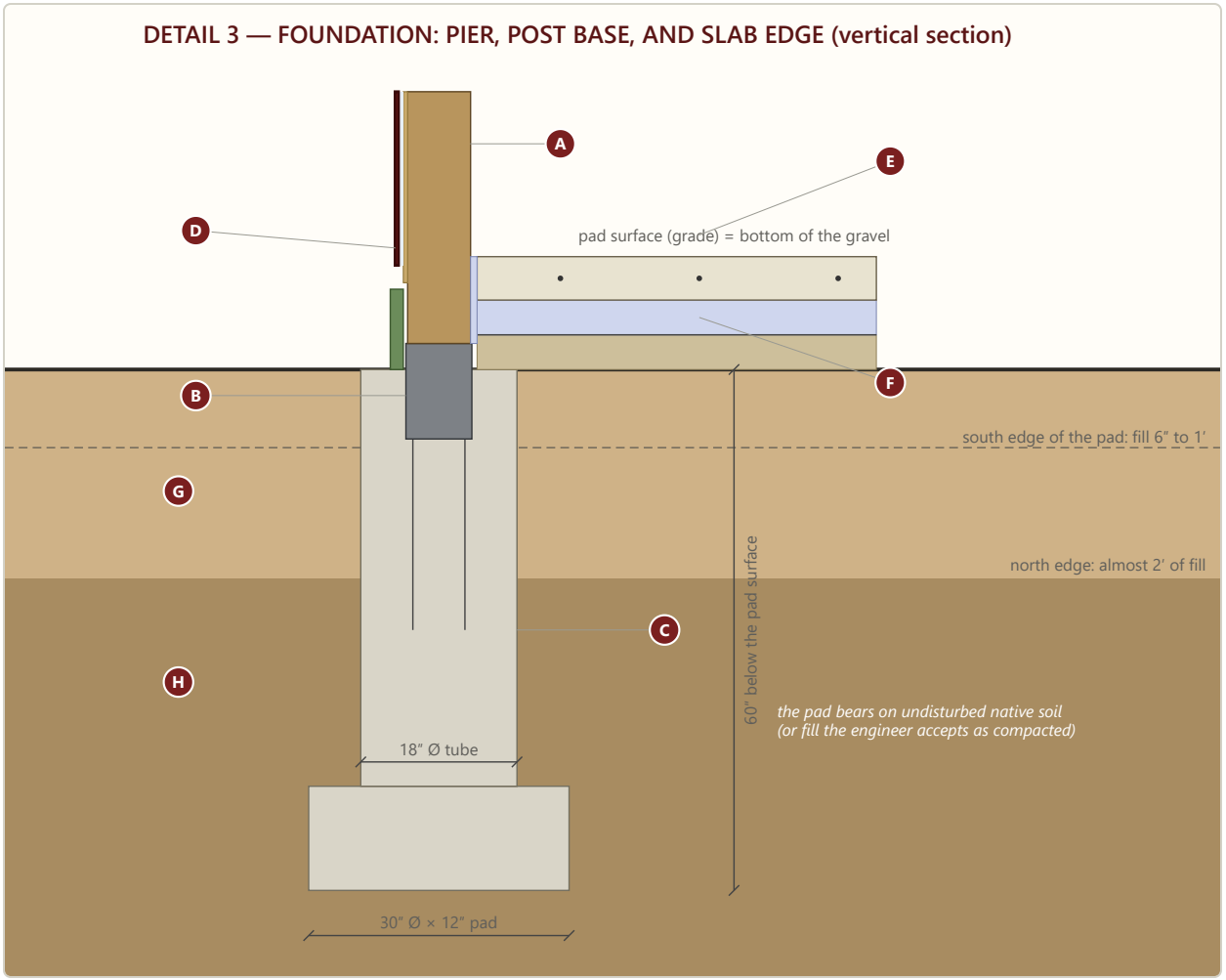


Layers are exaggerated. The lower slope is about 49° and gets full ice and water shield; the upper slope (24°) gets it only at the pitch break and around penetrations. Dripstop on the steel's underside controls condensation in the loft.

Marker	Layer
A	26 ga steel roof panel, factory Dripstop on the underside, beige, cut to the full slope length; #10 × 1½" hex screws with EPDM washers in the manufacturer's high-snow pattern
B	Synthetic underlayment (Titanium UDL30 or equal) over the entire roof
C	Ice and water shield: FULL on the lower (steep) slope; at the pitch break and around penetrations on the upper slope
D	⅝" OSB roof deck with H-clips between the purlins
E	2×6 SPF purlins on top of the truss chords, 16–24" OC per the truss design
F	Pre-engineered gambrel attic truss, 32' span, 2' OC, PE stamped for 120 psf ground snow

Detail 2, the stack-up on the lower (steep) gambrel slope, shown flat with the layers exaggerated. The real stack is under 4" thick, plus the truss chord. The snow guards are mounted separately over the walk door.

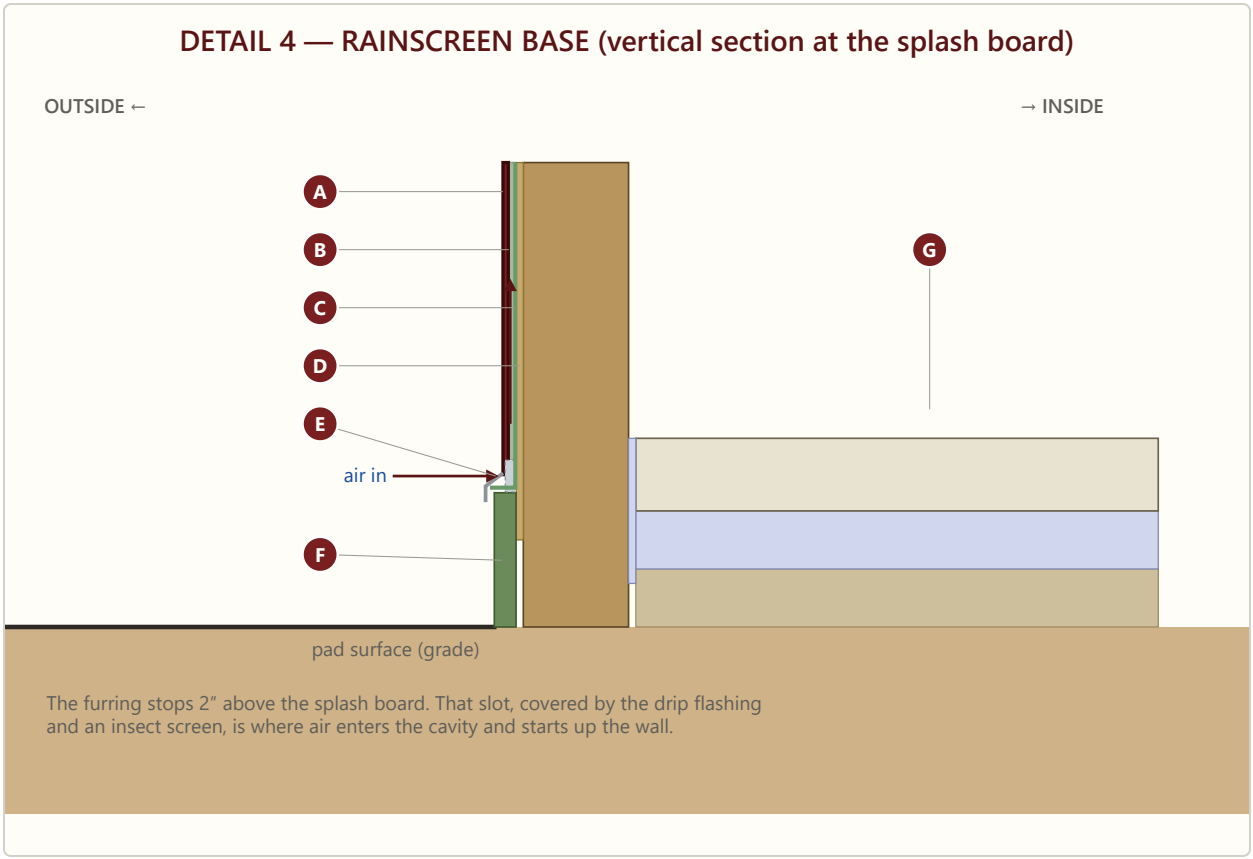
Detail 3: Foundation, Pier, and Post Base



Detail 3, a vertical section through a sidewall post, drawn at the north edge of the pad where the fill is thickest. The wet-set bracket lifts the post above the concrete, so wood never touches the ground. The pier passes through the imported fill and bears on native soil. The slab stands on its own, with an isolation strip against the post.

Marker	What it is
A	3-ply 2×8 laminated post (4½" × 7¼")
B	Perma-Column or Sturdi-Wall wet-set bracket, set in the wet pier concrete
C	4,000 psi concrete pier: an 18" tube on a 30" Ø × 12" pad, bottom 60" below the pad surface (the county's 30" below the frost line)
D	2×10 PT splash board on the outside of the post
E	5" slab, #4 rebar at 16" OC each way
F	4" rigid foam, 10 mil vapor barrier, and 4" compacted gravel, on the compacted fill
G	Imported fill, the level pad: up to about 2' at the north edge, 6" to 1' at the south edge
H	Native soil: the pier pad bears here, or on fill the engineer accepts

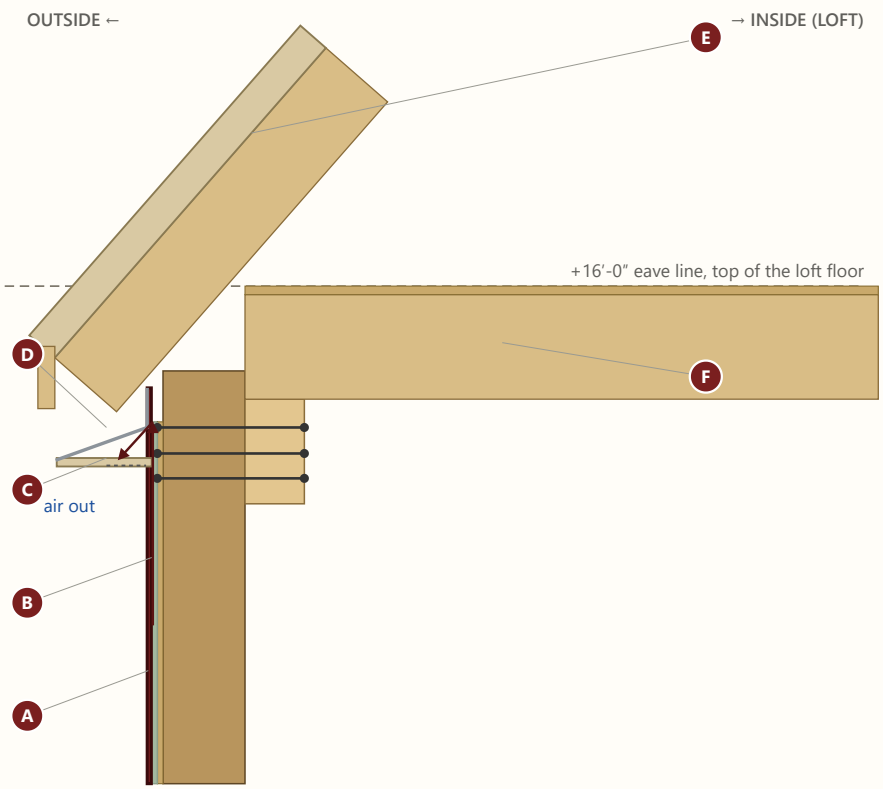
Detail 4: Rainscreen Base



Detail 4, the base of the wall where the rainscreen takes in air. The wrap shingle-laps over the top of the splash board. The furring stops short of the splash board, leaving a vent slot covered by the drip flashing and an insect screen. Air enters here, rises through the cavity, and leaves at the eave (Detail 5).

Marker	What it is
A	26 ga steel panel, bottom edge about 1" above the splash board's top
B	1×4 PT furring, stopping 2" above the splash board
C	Tyvek WRB, lapping over the top of the splash board, shingle style
D	½" OSB sheathing
E	Drip flashing and insect screen: the rainscreen's air inlet
F	2×10 PT splash board, proud of the steel, keeping the siding off the ground and the snow
G	5" concrete slab on foam and gravel

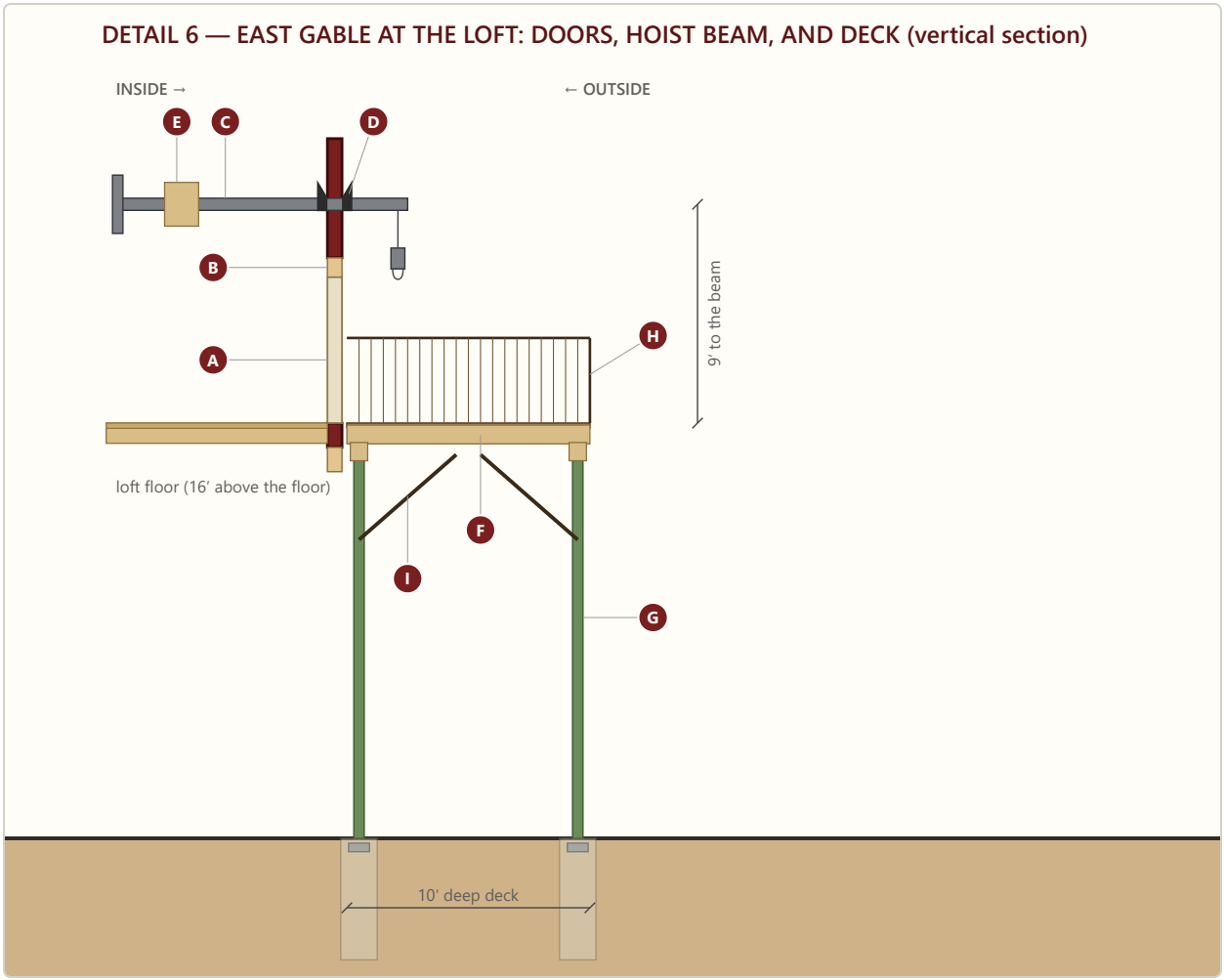
Detail 5: Eave (the Rainscreen Vents Out)



Marker	What it is
A	26 ga steel siding, up to the eave trim
B	1×4 PT furring, continuous to the soffit
C	Vented soffit closure: the rainscreen's air outlet
D	Eave trim: laps under the roof and over the wall, and wraps onto the sub-fascia
E	Roof assembly on the truss top chord: steel, underlayment, ice and water shield, OSB, and purlin
F	Truss bottom chord and ¾″ T&G loft floor

Detail 5, a vertical section where the rainscreen vents out at the eave. Air that entered at the splash board (Detail 4) rises through the furring cavity and leaves at the vented soffit closure. Without that top vent the cavity becomes a dead-air pocket that traps moisture.

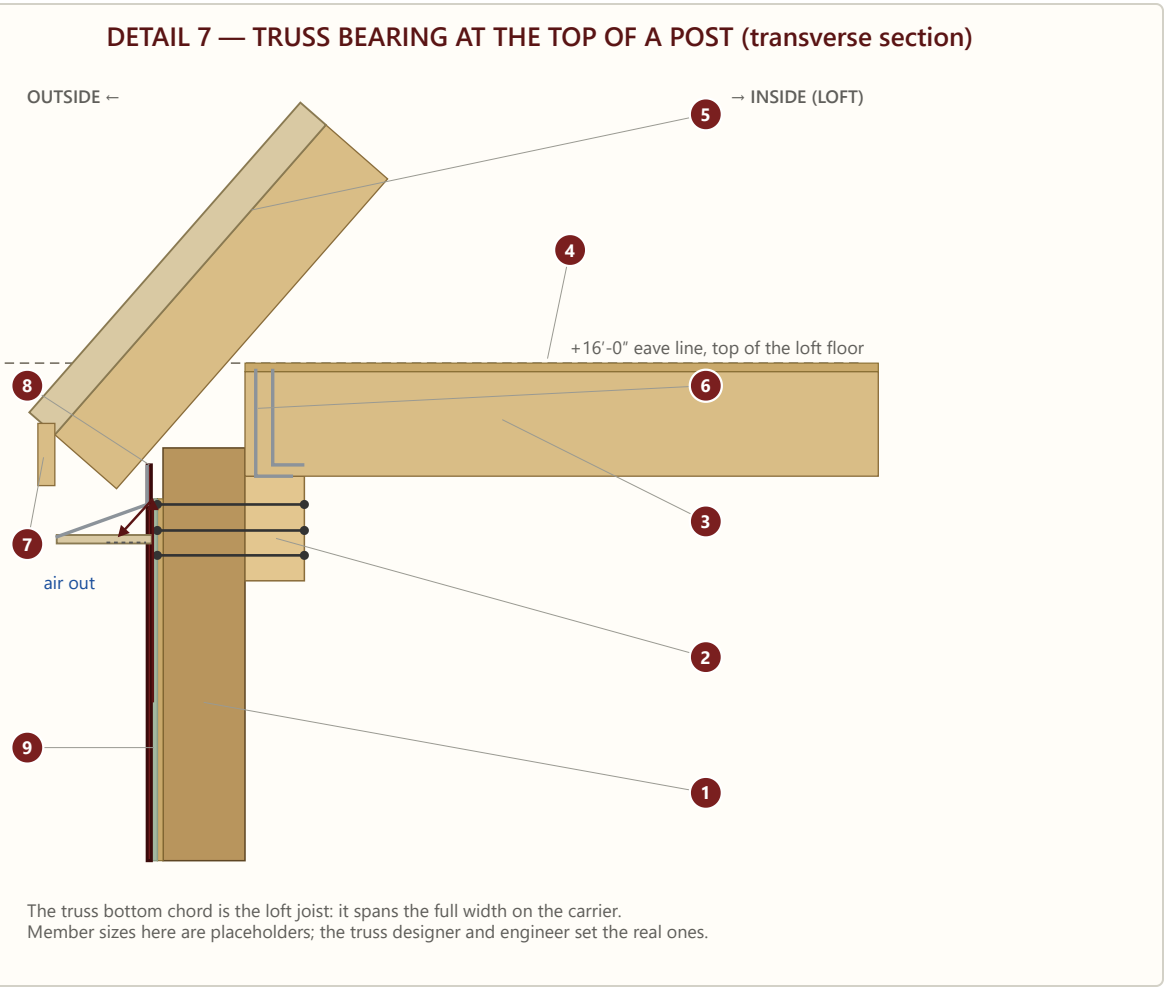
Detail 6: Loft Door, Hoist Beam, and Landing Deck



Detail 6, a vertical section through the east gable at the loft. The loft doors have an LVL header; the cantilevered W6 hoist beam passes through the wall and out over the deck, anchored to the truss top chord inside with a weather boot at the penetration; the free-standing 14′ × 10′ deck stands on 6×6 posts to pier footings, braced with diagonal knees.

Marker	What it is
A	5′ × 6′ insulated double doors at the loft floor, locking from inside, opening clear for the hoist
B	(2) 2×10 LVL header above the loft doors (engineer-sized)
C	W6×9 or W6×12 steel hoist beam, about 12 ft: about 9 ft inside and 3 ft out, 9′ above the loft floor, rated for 500–1,000 lb
D	EPDM weather boot and flashing where the beam passes through the wall: a CRITICAL detail
E	Anchor plate and bolt-through hardware to the truss top chord, with blocking between trusses to spread the load. The truss manufacturer reviews it
F	Deck framing: doubled 2×10 PT beams, 2×10 PT joists at 12″ OC, decking on top
G	6×6 PT posts on 18″ Ø piers, 60″ below grade, in post bases
H	42″ railing (fall protection above 30″)
I	Diagonal knee braces: a 16 ft tall deck will sway without them

Detail 7: Truss Bearing at the Top of a Post



Marker	What it is
1	3-ply 2×8 laminated post (4½″ × 7¼″) on a wet-set bracket below
2	(3) 1¾″ × 9¼″ LVL truss carrier, side-bolted to the post with (3) ½″ galvanized through-bolts per post
3	Truss bottom chord (2×10, a placeholder size), bearing on the carrier and acting as the loft joist across the span
4	¾″ T&G plywood loft floor, long edge across the chords
5	Truss top chord (2×8, a placeholder size), the lower gambrel slope rising to the knuckle
6	H2.5A hurricane ties, 2 per chord bearing, one on each side
7	2×6 sub-fascia closing the end of the truss tail (about 12″ of overhang) over the vented soffit
8	Wall steel stops at the eave; the metal trim wraps onto the sub-fascia
9	Wall assembly outside the post: steel, rainscreen, wrap, and OSB (Detail 1)

Detail 7, a transverse section through one sidewall post at the eave. The gambrel attic truss's bottom chord **is** the loft joist: it spans the full 32 ft with no bearing between the walls and lands on a **(3) 1¾″ × 9¼″ LVL truss carrier** side-bolted to the inside face of every post. The member sizes are placeholders.